Conference Management System

Technical Documentation – Swapna Swadhikta Behera

Backend & Frontend Contributions | Milestone 1 and Milestone 2

1. Overview

This document records Swapna's assigned responsibilities and evidenced contributions to the Conference Management System (CMS) prototype across Milestone 1 (backend) and Milestone 2 (frontend). It uses the supplied Git history, file-change log, branch list, current code archive, and updated team assignment as evidence.

The prototype uses FastAPI and SQLAlchemy for the backend and React + TypeScript (Vite) for the frontend. The backend endpoints are designed to be exercised through Swagger UI. The updated assignment identifies Swapna (Member D) as contributor of submissions, reviews, announcements, certificates, and search, with the camera-ready submission endpoint as her first Milestone 2 backend task.

Attribution note: A Git commit, merge, or file appearing in the final snapshot does not prove that every line was written by one person. This report distinguishes Swapna-authored commits from Snehansha's later integration/refactoring work and from other members' feature ownership.

2. Role and Assigned Scope

Milestone	Assigned responsibility	Files / deliverables
M1 – Backend	Submissions, reviews, announcements, certificates, search and filtering	backend/routers/submissions_router.py; reviews_router.py; announcements_router.py; certificates_router.py; search_router.py; related model/router registration integration
M2 – Backend addition	Camera-ready submission endpoint before frontend wiring	POST /submissions/{id}/camera-ready; submission status support for camera_ready_submitted and final_accepted where required
M2 – Frontend	Author submission UI, organizer submission management, review UI, search UI and reusable SearchBar	SubmissionsPage.tsx; OrganizerSubmissionPage.tsx; ReviewsPage.tsx; SearchPage.tsx (present in Git history but not in final archive listing); SearchBar.tsx

Shared integration	Resolve integration conflicts and make the frontend/backend run together	main.py routing/CORS integration; TypeScript conversion/UI integration; package/tsconfig/lockfile conflict resolution

3. Milestone 1 – Backend Contributions

3.1 submissions_router.py

Location: backend/routers/submissions_router.py

Purpose: manages the paper/abstract submission lifecycle for authors and organizers. This module connects paper submission with assignment, review, decision, and camera-ready stages.

- Creates submissions from author-provided paper information and associates the record with the authenticated author.
- Allows authors to retrieve their own submissions and supports organizer access to conference submissions.
- Supports organizer-controlled status updates used in the desk-check, review, decision, and final-acceptance workflow.
- Includes ownership and role checks for the camera-ready endpoint: the author must own the submission and the submission must be in the accepted state before camera-ready material can be submitted.
- The camera-ready action updates the workflow to camera_ready_submitted; the organizer can later move the paper to final_accepted through the status workflow.

Git evidence: commit 9b016658 (3 Sep 2026) introduced the submission feature set; commit 38588b4 (23 Sep 2026) explicitly added POST /submissions/{id}/camera-ready and role checks; commit 18be508 (26 Sep 2026) records TypeScript refactoring/UI integration and another update to submissions_router.py.

3.2 reviews_router.py

Location: backend/routers/reviews_router.py

Purpose: provides the reviewer assignment and evaluation workflow associated with a submission.

- Allows an organizer to assign a reviewer to a submission.
- Allows the assigned reviewer to submit a score, recommendation, and comments.
- Provides review retrieval for reviewer or organizer workflows, including reviews associated with a submission.

- Works with the submission status workflow so an organizer can make a final decision after review activity.

Git evidence: reviews_router.py was included in commit 9b016658 (3 Sep 2026), which explicitly names reviews among the new modules. The pull request merged on 4 Sep 2026 was titled “Integrated announcements, reviews, certificates, and submission feature modules.” Later file-change logs show Snehansha modified reviews_router.py during integration; those changes are recorded separately in Section 5.

3.3 announcements_router.py

Location: backend/routers/announcements_router.py

Purpose: allows organizers to publish conference announcements and users to retrieve announcements for a conference.

- Provides the backend operations for creating announcements and listing announcements by conference.
- Forms the backend service used by the Milestone 2 AnnouncementsPage UI, assigned to Pratyusha.

Git evidence: the file was introduced in commit 9b016658 (3 Sep 2026). Snehansha later made substantial edits to this router in the 4 Sep “Integrate backend router updates” commit; this is collaborative integration/refinement rather than a reason to attribute all current lines to Swapna.

3.4 certificates_router.py

Location: backend/routers/certificates_router.py

Purpose: supports certificate generation, user certificate listing, and public certificate verification.

- Generates certificates associated with a user and conference.
- Checks attendance eligibility before issuing a certificate in the current implementation.
- Prevents duplicate certificates for the same user/conference where an existing certificate is found.
- Provides public UUID-based verification without requiring the verifier to use the authenticated user flow.
- Provides an endpoint for the logged-in user to list their certificates.

Git evidence: certificates_router.py was introduced in commit 9b016658 (3 Sep 2026) and included in the 4 Sep integration pull request. Later changes by Snehansha are visible in the file-change log, so current behavior is described from the supplied source snapshot while ownership of individual later edits is qualified.

3.5 search_router.py

Location: backend/routers/search_router.py

Purpose: provides text search across the conference information supported by the prototype.

- Supports searching by query and type, with the assignment specifying conference, session, and speaker as searchable types.
- Uses the backend search endpoint that the SearchPage and SearchBar frontend components call.

Git evidence: search_router.py was introduced in commit 9b016658 (3 Sep 2026). Snehansha's 4 Sep integration update also modified the search router. The current source archive includes the backend router.

4. Milestone 2 – Frontend Contributions

4.1 SubmissionsPage.tsx

Location in Git history: frontend/frontend/src/pages/SubmissionsPage.tsx

Purpose: author-facing paper submission and tracking UI connected to the submission backend.

- Provides a form for submission details and sends the submission request to the backend.
- Lists the logged-in author's submissions and displays current status.
- Includes camera-ready submission interaction for an eligible accepted paper.
- Provides loading, success, and error feedback around API calls.

Git evidence: the file was added in the 26 Sep commit 18be508. The final ZIP contains SubmissionsPage.tsx, and the current code snapshot has the camera-ready backend endpoint available.

4.2 OrganizerSubmissionPage.tsx

Location: frontend/frontend/src/pages/OrganizerSubmissionPage.tsx

Purpose: organizer-side view for inspecting submissions and updating submission status as part of the review lifecycle.

- Provides organizer access to submission records and status controls.
- Complements the author-facing SubmissionsPage by separating organizer operations from author actions.

Git evidence: the page was added in the 26 Sep commit 18be508. In the updated assignment, the Content Management Page is assigned to Souptik and is a broader operational tracking page; the final archive lists OrganizerSubmissionPage.tsx rather than ContentManagementPage.tsx.

Therefore, the report does not claim that Swapna implemented the full ContentManagementPage.

4.3 ReviewsPage.tsx

Location in Git history: frontend/frontend/src/pages/ReviewsPage.tsx

Purpose: frontend for reviewers to inspect assigned papers and submit review scores, recommendations, and comments; organizers can also inspect submission reviews.

- Connects review form actions to the review backend.
- Presents review-related information for the reviewer and organizer workflow.
- Was added as part of the TypeScript/UI integration work on 26 Sep 2026.

Important attribution detail: the Git history shows Snehansha later modified ReviewsPage.tsx on the shared branch. The final source reflects collaborative iteration rather than a file untouched after Swapna's initial contribution.

4.4 SearchPage.tsx and SearchBar.tsx

Purpose: provide search controls and present matching conference-related results; SearchBar is intended to be reusable in the shared app layout.

- SearchPage.tsx was introduced in the 26 Sep TypeScript/UI integration history and subsequently received changes, including a search fix noted in Souptik's 27 Sep commit.
- SearchBar.tsx was added during the 26 Sep integration and later moved/modified across the component/page paths in Git history.
- The updated assignment states that Swapna should share SearchBar.tsx with Snehansha for placement in the Layout header.

Snapshot caveat: the final ZIP contains frontend/frontend/src/components/SearchBar.tsx and also a frontend/frontend/src/pages/SearchBar.tsx, but its file listing does not contain SearchPage.tsx. The Git history does show SearchPage.tsx at earlier points. This discrepancy is reported rather than silently assuming the page is present in the final snapshot.

5. Changes Made by Snehansha

Snehansha's role was Project Integration Lead. The supplied history shows both integration work across the repository and direct edits to files assigned to Swapna. The following points separate evidenced edits from general collaboration.

Area / files	Evidence from supplied Git records	Contribution description

Backend router integration: announcements, certificates, reviews, search, submissions	4 Sep 2026 commit f49ee34, “Integrate backend router updates,” lists all five routers as modified (705 insertions, 178 deletions across eight files).	Refined/integrated router implementations and aligned modules with the shared backend. These edits should be attributed as Snehansha’s integration/refinement, not as the initial authorship of every module.
main.py and models.py	4 Sep commit 9000ec4 “Update project implementation” modified main.py and models.py; later history also shows main.py/model changes during integration.	Adjusted shared routing/model structure while bringing modules together. This affects how Swapna’s routers are registered and how their data models fit the application.
reviews_router.py and certificates_router.py	Git file-change records show changes by Snehansha in later integration commits; the current history includes changes to reviews_router.py and certificates_router.py.	Made later compatibility and workflow adjustments in the shared backend. Specific line-by-line ownership should not be inferred solely from the final files.
submissions_router.py / models.py	Swapna’s camera-ready commit is 38588b4; later shared-branch history also records submissions_router.py/ model changes during integration and refactoring.	The camera-ready endpoint and role checks are directly evidenced in Swapna’s own commit. Later edits to the same files must be treated as collaborative follow-up rather than reassigned wholesale.
Shared frontend API/ authentication	20 Sep commits 90cb5c5 and 997f0e6: “Add shared API client with JWT authentication”; 29 Sep commit 152fd7b modified App.tsx, Layout.tsx, access.ts, api.ts, and multiple pages; 6 Oct history records a login fix.	Provided/updated the shared Axios client and JWT integration, app routing/layout/ access conventions, and authentication integration used by Swapna’s frontend pages.
Frontend integration and login	24 Sep commit 703605c added SponsorsPage, ExhibitorsPage and ForecastPage; 26 Sep commit 18be508 by Swapna records TypeScript refactoring/UI integration; 6 Oct Snehansha commits include LoginPage.tsx and auth.py changes.	Snehansha’s work connected pages into the overall app and refined login/authentication behavior. The work is separate from Swapna’s submission/ review/search feature ownership.

Branch synchronization and conflict resolution	Swapna commits 02df3ec (26 Sep) and 9a9e57a (26 Sep); merge commits 3640549 (4 Sep), a8b25ab (8 Oct), and 0030cac (8 Oct).	Swapna also performed integration work herself: resolving TypeScript/lockfile conflicts, updating main routing/CORS, restoring deleted review-related files, and merging Snehansha/main changes into her branch.

6. Cross-Team Contributions and Dependencies

Team member	Owned area in the assignment	Cross-contribution / dependency relevant to Swapna
Snehansha – Integration Lead	Backend foundation, database/ models/auth, shared API client, app shell/layout, sponsor/exhibitor and resource forecast pages, final integration.	Auth helpers and shared API client support authenticated submission/review actions. main.py/router registration and models are shared integration dependencies. Attendance records maintained in the shared backend are required for certificate eligibility. Snehansha also directly modified some Swapna-owned routers during integration.
Adrija – Member A	Auth/dashboard/feedback and reviewer workload balancer; frontend dashboard, feedback, reviewer workload, certificates UI per M2 assignment.	Reviewer workload analytics read submission/ review assignment data owned by Swapna but should not modify Swapna's tables. Dashboard counts and review workload summary depend on accurate submissions/reviews records. Certificate UI is a separate frontend page assigned to Adrija in M2, while certificate backend is Swapna's responsibility.
Pratyusha – Member B	Conference/session administration, scheduling, speaker assignment, room utilization optimizer; conference/session/room UI and announcements UI.	Submission, review, announcement, and certificate records require valid conference/ session identifiers. Pratyusha's conference/ session endpoints provide the real conference context; AnnouncementsPage consumes Swapna's announcements API.

Souptik – Member C	Registration, payment, attendance, bottleneck detector, content management page.	Attendance data is a direct input to Swapna's certificate eligibility check. The bottleneck detector reads submissions/review assignment state but should not modify Swapna's tables. ContentManagementPage consumes submission and review endpoints and depends on the camera-ready endpoint Swapna was assigned to finish first. The final ZIP does not list ContentManagementPage.tsx, so its completion is not confirmed by that snapshot.
Swapna – Member D	Submissions, reviews, announcements, certificates, search; camera-ready endpoint; submission/review/ search frontend.	Provides paper workflow and APIs used by organizer operations, reviewer workload analytics, bottleneck alerts, announcements UI, and certificate flow. Also participated in integration and conflict resolution.

7. Git Timeline – Key Evidence

Date / commit	Evidence and significance
3 Sep 2026 – 9b016658	"features added: submissions, reviews, announcements, search, certificates"; initial feature set and related backend files recorded as added.
3 Sep 2026 – 49e3c65	Created a separate routers folder and moved the five routers into backend/routers; this is repository organization work, not new feature logic by itself.
4 Sep 2026 – f5f94aa / 3640549	Pull request merged the submissions/reviews/announcements/ certificates feature set; Swapna's branch also merged Snehansha's branch.
4 Sep 2026 – f49ee34 / 9000ec4	Snehansha integration changes touched Swapna's routers plus shared main.py/models.py, establishing later collaborative refinement.
20 Sep 2026 – 90cb5c5 / 997f0e6	Snehansha added/updated shared API client with JWT authentication.
23 Sep 2026 – 38588b4	Swapna added POST /submissions/{id}/camera-ready with role checks.
26 Sep 2026 – 18be508	Swapna's commit records TypeScript refactoring/UI integration and submissions_router.py update; added App.tsx, main.tsx, OrganizerSubmissionPage.tsx, ReviewsPage.tsx, SearchPage.tsx and SubmissionsPage.tsx in the logged changes.
26 Sep 2026 – 02df3ec / 9a9e57a	Swapna resolved TypeScript/lockfile conflicts, updated main routing/CORS, and restored deleted files for review.

24 Sep–6 Oct 2026 – Snehansha commits	Snehansha integrated sponsor/exhibitor/forecast pages, updated app routing/layout/API/access logic, modified attendance and review/submission pages, and addressed a login issue.
8 Oct 2026 – a8b25ab / 0030cac	Swapna merged Snehansha's branch and origin/main into Swapna, reflecting final team synchronization.

8. Contribution Summary

Contribution area	Swapna's evidenced work	Shared / cross-team context
Submission backend	Initial submission feature; submission lifecycle; camera-ready endpoint with role/ownership checks.	Later integration edits to submissions_router.py/ models.py occurred on shared branches.
Review backend	Review assignment and evaluation feature integrated with submission workflow.	Snehansha later modified reviews_router.py; Adrija's workload balancer reads assignment data.
Announcements backend	Announcement create/list feature.	Pratyusha owns the announcement frontend; Snehansha later refined the router during integration.
Certificate backend	Generation, attendance eligibility, duplicate prevention and public UUID verification in current source.	Depends on Souptik's attendance records; Adrija owns the M2 Certificates UI per assignment.
Search backend/frontend	Search API and UI work; reusable SearchBar intended for app layout.	Snehansha integrates SearchBar into Layout; Git history records a search fix by Souptik and path changes/ renames.
Submission/review frontend	SubmissionsPage, OrganizerSubmissionPage, ReviewsPage, and Search UI artifacts evidenced in Git history.	Shared API client, JWT, app shell, routing, and some later page edits came from Snehansha and other contributors.
Integration	Conflict resolution, TypeScript conversion/UI wiring, main routing/CORS updates, deleted-file restoration, branch merges.	Work spans beyond Swapna's five owned modules and should be described as integration, not exclusive ownership of all integrated code.

9. Scope Differences and Evidence Limitations

- The updated team assignment is a work plan; it does not prove that every listed task was completed. This report only treats features as evidenced when supported by Git history or the supplied source archive.
- The final ZIP contains SubmissionsPage.tsx, OrganizerSubmissionPage.tsx, ReviewsPage.tsx, and SearchBar.tsx, but its file listing does not include SearchPage.tsx even though the Git history records it. SearchPage availability in the final snapshot is therefore uncertain.
- The updated assignment gives Souptik ownership of ContentManagementPage.tsx. The final ZIP listing does not show that file, so this report does not claim Swapna implemented the entire Content Management page.
- The current certificate router contains attendance validation and duplicate prevention; the updated assignment initially described temporarily skipping attendance checks and wiring them later. The report describes the supplied current source behavior, while recognizing the staged plan.
- Snehansha's commits modified several files originally assigned to Swapna. The report identifies those as integration/refinement work and avoids attributing every current line to either person without line-level blame evidence.
- Git commit messages and file-change summaries show contribution history, but they do not replace runtime testing. Swagger/browser demo completion should be recorded separately if supported by test results or a demo recording.

10. Reference Evidence Files

- git_history.txt – chronological commit list, authors, dates, and merge messages.
- git_detailed_history.txt – commit metadata and file-level change records.
- git_file_changes.txt – changed files and insertion/deletion summaries for commits.
- git_branches.txt – branch and remote-tracking branch names.
- Conference_System_History.bundle – repository history bundle supplied with the project.
- Conference_System_Code.zip – final source snapshot dated 8 Oct 2026.

End of report.